

ows and dirt on even a part of a panel reduced sharply the energy output of not only that panel but the entire string. This shadow intolerance of the string led to drastically reduced energy output of the entire string.

3. Currently deployed series+parallel configuration of the solar array is fed into the MPPT current controller provided by the inverter supplier. Such MPPT looks for extracting peak power from the array as a whole. This process of averaging does not ensure that each panel is delivering its peak power. This also leads to not using each panel to contribute its maximum possible energy.

4. Since the solar output per panel is limited to 250Wp to 300Wp, one needed several series strings of panels in parallel. The matrix and its maintenance thus became complex and expensive.

ESPs overcome the above shortcomings

1. Since ESPs provide stable high-voltage output and have integrated DSP-based MPPT controller, even hundred or more ESPs in an array can now be hooked up in parallel. Constant high-voltage DC output from ESPs also eliminates voltage swings that one encounters in arrays built with series-connected normal PV panels where output voltage swings over 100 per cent under various weather conditions.

2. It is here that the shade-tolerant characteristics of the parallel array of ESPs come into their own; the ESP array has a multitude of panels in parallel feeding a single bus bar. This avoids dramatic loss in the power output of presently used conventional arrays. Shadow on one ESP in parallel architecture just affects the power contributed by that particular panel. In an array of hundreds of panels, this is a minor loss. If one or two are in shadow or dirt covered, the rest keep functioning.

3. DSP-based energy controller embedded in ESPs ensures that each panel in an array delivers peak energy output from its incident solar input. This gives considerable energy gain

since the incident solar energy can vary widely over the large array fields with hundreds of panels.

4. Each ESP output current is small due to its high output voltage. This reduces the current per panel to less than one ampere, and thus the cost of inter-connections. One single high-current bus bar delivers power to the inverter or the battery with minimal losses, maximum simplicity and much lower cost.

The ultimate measure of the solar array performance is the AC power that is finally produced by the array-fed power inverter. Problems in current limitation due to series and parallel configuration result in loss of energy harvesting as well as complexities due to panel variance. In addition, direct shading and other factors such as soiling, panel degradation or thermal variation across panels are the causes for inefficient power generation. All these problems are effectively addressed by the ESP technology. As ESPs embed maximum power-point tracking right at the panel, any underperforming ESP does not bring down the overall performance. By using ESPs, each panel maintains peak performance regardless of the conditions affecting it or the other panels around it.

Converting an inverter or UPS into a solar system

ESPs' built-in DSP-based energy control and energy monitoring as well as high output voltage has one major advantage: When one uses ESP arrays, the solar inverters fed by these arrays do not need any built-in high-power MPPT current controller. This reduces the cost of the inverter drastically. It also reduces stress on the inverter's critical components, eventually increasing their lifetime and aiding them to operate at higher efficiency.

The high DC bus voltage can be fixed at the optimal level for any inverter, according to its specifications, improving inverter efficiency and reliability. In combination with distributed MPPT, this enables solar photovoltaic systems

to harvest 10 to 30 per cent more energy than conventional systems.

Panel performance monitoring

ESPs allow panel-level monitoring for enhanced visibility into system operation to help with maintenance and ensure long-term system performance. This simultaneously addresses fundamental problems facing plant designers, installers and operators and provides quantifiable incremental performance without increasing the cost.

ESPs are environmentally rugged since the embedded system operates without deterioration at ambient temperatures up to 80°C.

Lower system cost

Use of ESPs lowers the installed cost and improves the performance of solar power systems, while also eliminating the design and installation constraints inherent in conventional string architectures. The parallel solar architecture enables connection of over 100 ESPs on each cable run; twenty-fold improvement over conventional string architecture. This results in significant installed-cost savings.

ESPs also mitigate the effects of panel mismatch due to shading, soiling and manufacturing variations, which can easily disrupt the performance of an entire string array. This enables solar photovoltaic systems to harvest up to 30 per cent more energy than conventional systems.

Designers can focus their efforts on power production rather than component interaction. Simplified wiring, easy interconnection, unit-level monitoring and reduced inverter expenses assure operators of long-term performance and lower costs throughout the system's life-cycle. ●

MITRAMAX Energy was set up by Mr Deodhar after he stepped down from his responsibilities with Apla as chairman. Mr Deodhar has devoted himself to research in solar energy harvesting technologies and ESP is its first product for which he has patent pending. He can be reached at p_deodhar@hotmail.com